

Journal 05: Intellectual Property Risks in AI-Generated Code

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The recent explosion of artificial intelligence (AI) has rippled its way through almost every sector of day-to-day lives, from customer service and social media content creation to application development and business analytics. However, there are many intellectual property (IP) risks associated with using AI that people are only now beginning to consider. First this paper will briefly explain the difference between copyright and licensing in the context of AI-generated code. An enumeration of the common IP risks such as the question of ownership, licensing conflicts, and copyright infringement will follow. Finally, a discussion on developer best practices for protecting IP will provide direction for future work.

While copyright and licensing are often lumped together, they are subtly different. Copyright encompasses the IP rights that an author intrinsically has after the creation of a creative work. Nothing needs to be done to copyright something, it happens automatically. A license is a ruleset that an author provides for utilizing their creation in derivative works. There are many types of licenses, but the two most common are permissive and copyleft licences. Permissive licenses grant others the right to use the content in any way they please, without the need to redistribute the license (Black, 2023). On the other hand, copyleft licenses require that any derivative work carry the same—or more restrictive—license (Black, 2023). Apache is the most common permissive license while the GNU General Public License (GPL) is the prevalent copyleft license. Licensing promotes software freedom and collaboration and is an important cornerstone of the open-source community.

IP is not typically on the forefront of peoples' minds when they use AI, but it should be, especially for companies or anyone creating a product. AI models use IP in their training data and may also output content that is subject to IP protections (Lucchi, 2023). This raises the

question: is AI-generated content copyrightable, and who claims ownership over that content? In general, it seems that the consensus is yes, AI-generated content can be copyrightable, especially when a human prompter applies significant prompting, editing, or modification (Herbold et al., 2024; Lucchi, 2023). In these circumstances, the human interacting with the AI tool claims ownership of the work, because they are the entity providing the “creative work” that copyright requires (Lucchi, 2023). The AI tool itself cannot claim copyright for a few reasons. First, AI output is algorithmically determined (Lucchi, 2023). Second, United States copyright law states that work moves into the public domain 70 years after the death of the author. Obviously, a model does not “die” in the traditional sense, so the law would need to be modified before it could apply to AI. One way to think about copyright in the context of AI is to look at how cameras are handled. A photograph is copyrightable by the photographer, even though technically the camera is producing the actual image. In this scenario, the photographer utilizes the camera as a tool to create their work, just as a prompter uses a model (Lucchi, 2023).

Beyond the question of IP and copyright ownership, there are also licensing risks to consider in regard to AI-generated content. Models are trained on vast amounts of open-source and public information, which includes both copyrighted and licensed content. It is common for models to “memorize” aspects of their training data and output them in responses, often with no indication that said information is protected under IP rights (Herbold et al., 2024). This can lead to unintentional copyright or licensing violations. For example, Xu et al. (2025) show that AI is capable of producing “strikingly similar” code output from copyleft licensed libraries. In nearly all of these cases, the tool was also *incapable* of providing accurate information about the license status of the code. This means that even prompting models to be aware of licensing is not always safe (Xu et al., 2025). A major concern with using content under strong copyleft licenses is that

the entirety of the derivative work must be shared (Black, 2023; Herbold et al., 2024). For companies and enterprises, this would likely result in a destruction of their business model, as revenue typically relies on retaining some level of a product (Lucchi, 2023).

So, if licensing and copyright risks are such a large concern, how do developers avoid them? Currently, there are no perfect mechanisms for detecting IP and it seems that most have taken the approach to ignore the problem completely (Herbold et al., 2024). However, awareness of content protection and active practices can mitigate licensing violations. To begin, utilize any tools that the AI model provides. For example, Github's Copilot has a "duplication detection" feature that will attempt to detect copied code in any public repository (Black, 2023; Github, 2026). Ensure that all AI-generated code is tagged appropriately in each project, so that, if necessary, it is easier to refactor and remove code in violation (Black, 2023). Do not rely on the AI tool to provide accurate information about the licensing status of its generated content (Xu et al., 2025). Instead, perform an independent investigation in addition to utilizing external post-processing tools that scan for licensed content (Black, 2023). Despite the risks, AI is a powerful utility that should not be overlooked out of fear.

References

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